Ten Important Skills: Sample Quizzes

1. a) Let A be the following complex matrix and let v be the following vector in $\mathbb{C}^3$. Find Av.

$$A = \begin{pmatrix} 0 & i & 1 \\ 1+i & 1 & 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}$$

b) Let B be the following real matrix:

$$B = \begin{pmatrix} 8 & 0 \\ 0 & 1 \\ 0 & 4 \end{pmatrix}$$

Complete the following sentence by filling in the four blanks:

The matrix B is a ___ $\times$ ___ matrix and it represents a linear transformation from $\mathbb{R}^{\_\_}$ to $\mathbb{R}^{\_\_}$.

2. a) For each of the following matrix operation, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$B - A$, BA, $B + C$, AC, BD

b) Give an example of two nonzero 2×2 matrices whose product is a zero matrix, or explain why this is not possible.

3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

rotate about the origin by $\pi/6$ counterclockwise and then reflect across the y-axis

b) Describe in words the linear transformation given by the following matrix, which uses the standard basis for $\mathbb{R}^4$.

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

4. a) Use row reduction to put the following matrix in reduced row echelon form.

$$\begin{pmatrix} 2 & 0 & -6 & -8 \\ 0 & 1 & 2 & 3 \\ 3 & 6 & -2 & -4 \end{pmatrix}$$

b) Give the general solution to the following linear system in vector form.

$$x_1 + 2x_2 + 3x_3 = 4$$
$$-x_1 + x_2 + 2x_3 = 5$$

5. a) Compute the inverse of the following matrix.

$$\begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

b) Let A and B be the following matrices. Compute the inverse matrix of the product AB.

$$A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

6. a) Give a basis for the column space of A.

$$A = \begin{pmatrix} 6 & 2 & 2 & 0 \\ 1 & 3 & 11 & 8 \\ 1 & 4 & 15 & 11 \end{pmatrix}$$

b) Give an example of a basis for $\mathbb{R}^3$ containing the vectors $(1, 2, 3)^T$ and $(4, 5, 6)^T$ or explain why this is not possible.

7. a) Using the standard basis, a linear transformation is given by the matrix A. Write the same linear transformation using v_1 and v_2 as an ordered basis for $\mathbb{R}^2$.

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}, \qquad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

b) Give an example of a pair of inverse matrices that are similar to each other.

8. a) Compute the determinant of this matrix:

$$\begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

b) Give two examples of a 3×3 matrix with determinant 14.

—Midterm exam—

9. a) Find the eigenvalues and eigenvectors of this matrix:

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

b) Is v an eigenvector of A? Explain why or why not.

$$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

10. a) Compute an orthonormal basis for the subspace spanned by this set of vectors in $\mathbb{R}^3$:

$$\left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}$$

b) What are two vectors that are orthogonal to v?

$$v = \begin{pmatrix} 2 \\ 1 \\ 3 \\ -4 \end{pmatrix}$$